

FALSIFICATION-BASED PEDAGOGY

A Formal Framework for Teaching Through Error Elimination
Toward a Popperian Methodology of Instruction, Assessment, and Classroom Design

Gia Bao Huynh
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Abstract

Karl Popper's philosophy of science has been applied to education extensively at two levels: as a civic philosophy of the open society cultivating dispositional critical thinking (Lam 2007; Lam 2013; Zecha 1999), and as a narrow instructional technique within science curricula, where students generate and test hypotheses against discrepant evidence (Posner et al. 1982; Kang, Scharmann, and Noh 2004). Neither level formalizes falsification as the organizing principle of an entire pedagogical methodology — a system in which every classroom activity, assessment instrument, and teacher intervention is designed around students' capacity to generate the conditions under which their own beliefs would be shown false, rather than around their capacity to justify beliefs already held. This paper develops that formalization. It states five axioms derived from Popperian epistemology, translates them into a five-principle pedagogical framework, specifies classroom procedures and assessment criteria that operationalize each principle, and locates the resulting methodology against four established pedagogical traditions — inquiry-based learning, problem-based learning, the Socratic method, and Toulmin-style argumentation pedagogy — as well as three closely related but structurally distinct research programs: Taking Children Seriously, Manu Kapur's Productive Failure, and the cognitive-conflict tradition in conceptual-change research. The paper argues that what distinguishes Falsification-Based Pedagogy from each of these is not its philosophical premises, which are largely shared, but a specific and previously unformalized design choice: that the ability to generate a successful refutation of one's own hypothesis is treated as the primary object of instruction and assessment, not as an instrumental step toward a separately graded correct answer.

Keywords: Karl Popper; falsificationism; critical rationalism; pedagogy; assessment design; error elimination; conceptual change; productive failure

I. Introduction: What Popper Has Not Yet Been Asked to Do

Karl Popper's account of scientific method rests on an asymmetry that has no analogue in most classroom assessment: a theory can never be confirmed, only tested and, if it survives testing, corroborated; a single well-established counterexample, by contrast, is sufficient to refute it (Popper 1959). Confirming instances are cheap and, on Popper's account, epistemically close to worthless, since a sufficiently vague or ad hoc theory can always be made to fit whatever evidence a confirming instance supplies. What does epistemic work, on this account, is the deliberate, disciplined search for the observation, argument, or counterexample that a theory could not survive.

Most classroom assessment inverts this asymmetry. A student's grade typically tracks their ability to produce a confirming instance of the correct answer — to solve the problem, cite the correct rule,

reproduce the accepted explanation — and rarely tracks, as a graded competency in its own right, their ability to state the specific conditions under which the rule, explanation, or their own prior answer would be wrong. This paper asks what a classroom would look like if that asymmetry were reversed: if the object of instruction, and the object of assessment, were built around students' capacity to search for and construct refutations, with confirmation-seeking treated as the secondary, instrumentally necessary but epistemically weaker activity Popper's own philosophy of science treats it as.

This is not, on inspection, an unprecedented question. It has been asked, and answered, at two distinct levels, neither of which reaches the level this paper is concerned with, and it has been approached from three further, closely adjacent directions that require careful differentiation rather than a claim of unqualified novelty. Section II surveys all five before Section III begins the formalization proper.

II. What Already Exists

II.1 The Civic and Dispositional Level

The most extensive body of Popper-and-education scholarship treats critical rationalism as the epistemic precondition for Popper's open society (Popper 1945), and asks what kind of education such a society requires of its members: tolerance of disagreement, a disposition toward rational criticism, and resistance to what Popper called the bucket theory of mind, in which knowledge is transmitted rather than actively constructed. Lam's 2013 paper draws the connection between open-society values and their educational implications directly, and his 2007 paper asks explicitly whether Popper's falsificationist heuristic can serve critical-thinking pedagogy, answering largely in the affirmative at the level of general disposition: students should hold beliefs tentatively, welcome criticism, and actively resist confirmation bias (Lam 2007). Zecha's edited volume *Critical Rationalism and Educational Discourse* (1999) gathers scholars developing Popper's implications for moral education, political education, and educational research methodology itself. Joseph Agassi, Popper's most direct disciple to write extensively on education, developed a related but distinct thread across four decades of essays collected as *The Hazard Called Education*: not falsification as classroom method, but Socratic teaching as a disposition, and as the antidote to institutional authoritarianism specifically. None of this literature specifies a classroom procedure, an assessment rubric, or a design principle for a given lesson; it operates at the level of what kind of person, or what kind of political culture, Popperian epistemology should produce.

II.2 The Narrow, Science-Specific Level

A second, distinct literature treats hypothesis generation and testing as a specific technique within science instruction, closely related to but rarely citing Popper directly. The Conceptual Change Model formalized by Posner, Strike, Hewson, and Gertzog (1982) proposes that a misconception is replaced only when a student becomes dissatisfied with it, typically by confronting a discrepant event the misconception cannot explain — a mechanism subsequent research renamed cognitive conflict (Kang, Scharmann, and Noh 2004) and has studied extensively since; a 2024 meta-analysis of 218 primary studies found a large aggregate effect (Hedges' $g = 1.10$) for conceptual-change strategies built on this mechanism (Pacaci et al. 2024). This tradition is grounded explicitly in Piagetian equilibration and Festinger's cognitive dissonance theory, not in Popperian

epistemology, though the structural resemblance — a belief is revised because a specific instance defeats it, not because a teacher pours in a replacement — is close. Separately, Toulmin's Argument Pattern (Toulmin 1958), imported into science education by Driver, Newton, and Osborne (2000) and in wide use since (Erduran 2018), formally includes rebuttal — description and rebuttal of counterexamples — as one of six components students are trained to construct. But the Toulmin tradition trains students to build one complete, defensible argument, including anticipated objections to it; it does not reward students specifically for the act of defeating their own claim, and it does not treat refutation-generation as a graded competency independent of the argument's overall soundness.

II.3 The Closest Adjacent Frameworks, and Why Each Still Falls Short

Three further research programs deserve separate treatment because they come closer than either of the above to what this paper formalizes, while each stopping short of it for a specific, identifiable reason.

David Deutsch and Sarah Fitz-Claridge's *Taking Children Seriously* is the most directly Popperian of the three: it grounds itself explicitly in Popper's distinction between the bucket theory and the searchlight theory of mind, and in his account of knowledge growth through conjecture and refutation (*Taking Children Seriously*, *Unschooling* and *Karl Popper*). But its organizing principle is non-coercion, not falsification-as-assessment: it holds that any externally imposed evaluative structure, including grading, risks becoming what Popper called unwanted answers to unasked questions, interfering with the child's own rational, self-directed error correction (*Taking Children Seriously*, *Coercive Education*). A pedagogy that rewards successful refutation, of the kind formalized in Principle 3 below, is exactly the kind of extrinsically imposed evaluative structure *Taking Children Seriously*'s own premises treat with suspicion. The two programs share an epistemology and diverge sharply on institutional design.

Manu Kapur's *Productive Failure* is the most empirically substantiated of the three, tested across 166 experimental comparisons involving more than 12,000 participants (Sinha & Kapur 2021) and organized around a specific instructional sequence: students first attempt to generate solution methods for a complex, novel problem, typically failing to produce the canonical method, before receiving direct instruction that consolidates their attempts into the correct concept. *Productive Failure* is not explicitly grounded in Popperian epistemology, and its unit of analysis is a well-posed problem with a determinate correct solution method, not a propositional claim the student holds and is trained to actively falsify; the failure it designs for is failure to arrive at an answer, not success at refuting one.

The cognitive-conflict and conceptual-change tradition surveyed in II.2 is the closest of the three in mechanism — a specific counterexample or discrepant event defeats a specific belief — but the counterexample in this tradition is standardly supplied by the teacher or the demonstration apparatus, not generated by the student as a graded act of intellectual labor; the student's task is to notice and resolve a conflict the teacher has engineered, not to search for the conflict independently.

A fourth adjacent program, further from the classroom but structurally close to Principle 2 below, deserves inclusion for the same reason. William McGuire's inoculation theory (McGuire 1961; McGuire 1964) proposes that resistance to persuasion is built by pre-exposing a person to a

weakened version of a persuasive attack together with its refutation, on direct analogy with biological vaccination; Huynh's prior work connecting this literature to Popper (Falsifiability as Cognitive Self-Vaccination) links McGuire's mechanism to Popper's concept of immunizing stratagems — the structural property, common to propaganda and pseudoscience alike, by which a claim absorbs any potential counterevidence as consistent with itself — and argues that the falsifiability criterion, taught as a single structural test rather than a catalog of named techniques, is the most durable and general form inoculation can take. This is the closest existing literature to Principle 2's counterexample-generation requirement in its underlying logic, and it supplies a vocabulary this paper adopts directly: a claim is self-sealing when no stated observation could count against it, and training in falsifiability is training in detecting self-sealing structure regardless of its surface content. It falls short of formalized classroom pedagogy for the same reason fact-based and technique-based prebunking do more generally: inoculation content is standardly administered by a researcher, platform, or institution to a comparatively passive recipient, whereas Principle 2 requires the student to construct the counterexample as their own graded work, across ordinary subject-matter claims rather than against a pre-identified manipulation attempt.

None of these four programs, individually or combined, states a classroom methodology in which the primary graded activity is students' own construction of falsification conditions and counterexamples against their own and their peers' claims, across domains, as an assessed competency comparable in status to producing a correct answer. That is the specific, narrow gap this paper addresses.

III. Five Axioms

The framework rests on five axioms, each directly derived from Popperian epistemology and stated in a form suitable for translation into classroom design.

A1 — Asymmetry of Confirmation and Refutation. No finite number of confirming instances establishes the truth of a general claim; a single well-established counterexample is sufficient to establish its falsity (Popper 1959). Instructional and assessment design should therefore weight refutation-finding at least as heavily as confirmation-producing, contrary to the implicit weighting of most existing assessment instruments.

A2 — Fallibilism as Starting Condition. Every claim a student or teacher holds, including claims presented as settled curricular content, is provisionally true at best and should be presented and assessed as such, not as authoritative pronouncement — knowledge, on this axiom, remains forever conjectural (Popper 1963).

A3 — Non-Justificationism. The goal of rational inquiry is not to justify a belief from foundations that themselves require no justification — an infinite regress critical rationalism explicitly rejects — but to expose the belief to criticism and see whether it survives (Popper 1983; Lam 2013). Classroom activity organized around “justify your answer” therefore targets the wrong epistemic operation; classroom activity should instead be organized around “expose your answer to the strongest criticism available and report what survives.”

A4 — Learning as Error Elimination, Not Content Transfer. Knowledge does not grow by being poured into a receptive mind — the bucket theory, which Popper rejects — but by an active, conjectural process in which existing conjectures are tested against experience and error and

eliminated: the searchlight theory (Popper 1972). Instruction organized around content transfer is, on this axiom, not merely suboptimal but a category error about what learning is.

A5 — Separability of Person and Proposition. A refuted hypothesis is a success of the inquiry process, not a failure of the person who held it. This is not a Popperian claim per se, but it is a necessary institutional precondition for A1 through A4 to be implementable in a setting, like a classroom, where the person holding the hypothesis and the person whose performance is evaluated are the same individual. Without A5, rewarding refutation under A1 creates a direct incentive conflict with students' well-documented tendency toward ego-protective, confirmation-seeking behavior under evaluative threat — a tendency the growth-mindset literature documents from a different theoretical starting point but describes in compatible terms (Dweck 2006).

IV. The Five Principles

The five axioms translate directly into five operating principles for classroom design.

Principle 1 — Every Claim Carries a Falsification Condition (from A1, A3). No claim is accepted as a contribution to classroom discourse, from either student or teacher, until it is accompanied by a stated answer to: “What observation, argument, or counterexample would show this is wrong?” A claim for which no answer can be given is treated not as more secure but as less admissible than a claim with a stated, specific falsification condition, since an unfalsifiable claim has, on A1, forfeited the only mechanism by which its epistemic status could improve.

This principle carries an extension that the classroom setting makes easy to overlook: the test must be applied symmetrically, to claims a student is inclined to agree with as readily as to claims a student is inclined to doubt. Huynh's Principle of Epistemic Symmetry (developed for adult discourse in Falsifiability as Cognitive Self-Vaccination) states that a critic invoking falsifiability against a claim may not, in the same discourse, exempt claims from sources they trust from the identical standard. Applied to Principle 1, this means the falsification-condition requirement binds a textbook's claim and a popular claim the class already believes exactly as it binds a claim the class is skeptical of; a classroom that reliably demands falsification conditions from unfamiliar or minority positions while waiving the demand for familiar or majority ones has not implemented Principle 1. It has implemented confirmation bias with an unfalsifiability requirement attached to one side only.

Principle 2 — Students Generate Counterexamples (from A1, A4). The standard classroom task of finding confirming instances — find another example of X — is replaced, as the primary graded task, with its inverse: given a claim, find the one observation that would destroy it. This inverts the incentive structure of the cognitive-conflict tradition surveyed in §II.2, where the disconfirming instance is teacher-supplied; here, locating or constructing it is the student's own assessed work.

Principle 3 — Successful Refutation Is Rewarded (from A1). Assessment rubrics award credit, explicitly and at a weighting comparable to correct-answer production, for: identifying a valid counterexample to a stated claim, including the teacher's own or the textbook's; identifying a hidden assumption a claim depends on; identifying a boundary condition under which an otherwise correct claim fails; and identifying a logical inconsistency between two claims the class or the curriculum has both accepted.

Principle 4 — Identity Is Separated from Hypothesis (from A5). Classroom language and assessment feedback are structured to make “my hypothesis was refuted” and “I performed well” compatible rather than opposed statements — for instance, by grading the quality and validity of a self-generated refutation of one's own prior claim as a distinct, positively weighted category from the grade attached to the original claim, so that a strong refutation of a weak original answer can outscore an undefended correct guess.

Principle 5 — Revision Is the Success Condition (from A2, A4). The unit graded as complete is not the original answer but the revision cycle: claim, stated falsification condition, attempted refutation, and revised claim. A cycle that ends in revision following a successful refutation is scored as a complete success; a cycle that ends in an unrevised, unrefuted original claim is scored as incomplete, regardless of whether that original claim happened to be correct, since on A2 no claim's correctness is available to the classroom as a settled fact independent of the process that tested it.

V. Classroom Procedures

A unit organized under this framework proceeds through four stages, repeated at whatever grain size the subject matter and student age make workable — a single claim within a forty-minute lesson, or a multi-week unit built around one central conjecture.

Stage 1 — Conjecture. Students, or the teacher for foundational content the class does not yet have the tools to generate independently, state a claim in falsifiable form. A claim not yet in falsifiable form (“photosynthesis is important”) is first reformulated, as a class exercise, into one that is (“plants deprived of light for more than seventy-two hours will show measurable chlorophyll loss”) before the cycle proceeds.

Stage 2 — Falsification Condition. Before any testing, evidence-gathering, or argument occurs, students state what would count as a refutation. This step is graded independently of what follows: a well-specified falsification condition attached to a claim that later survives testing is scored on the quality of the condition, not on the outcome, since a well-specified condition the claim happens to survive has still done its epistemic job. What counts as well-specified is given a precise standard in §VI below.

Stage 3 — Refutation Attempt. Students, individually or in pairs assigned specifically to the adversarial role, search for the counterexample, hidden assumption, boundary case, or inconsistency specified in Stage 2. Where the domain allows direct empirical testing, this stage includes running the test the falsification condition specifies. Where it does not, this stage is an argumentative search structurally comparable to Toulmin's rebuttal construction (§II.2), but undertaken as an attack on a claim not one's own, or on one's own prior claim specifically, rather than as a defensive component of a single self-authored argument.

Stage 4 — Revision and Report. The claim is revised, abandoned, or narrowed in scope in light of Stage 3, and the class or student reports what happened to the claim, not only what the claim now says. A report of “the claim survived this specific, serious attempt to refute it” is treated as substantively different from, and initially weaker than, “no serious attempt to refute the claim has yet been made” — Popper's corroboration, which increases with the severity and number of failed refutation attempts a claim has survived, rather than confirmation, which does not (Popper 1959).

A refuted claim is not simply discarded. Huynh's *The Pre-Registered Afterlife* distinguishes four functional states a falsified body of work can occupy — historical artifact, counterfactual benchmark, and surviving conceptual repertoire, alongside the live explanatory status a claim loses once refuted — and argues that which state a falsified theory lands in is normally discovered by accident, long after the fact, rather than designed by the theorist at the time. A classroom can design it instead. Stage 4's report should specify, for any claim that does not survive Stage 3, which of three archival roles it will occupy: retained in a class record as a documented example of a well-constructed but ultimately incorrect hypothesis, useful for calibrating what good falsifiable reasoning looks like independent of whether the reasoning was right (the historical-artifact role); retained as a marked point of comparison against the revised claim that replaces it, so that the specific respect in which reality diverged from the original conjecture remains legible rather than silently overwritten (the counterfactual-benchmark role); or mined for the sub-components — a valid partial argument, a correctly identified variable, a working method — that survive the larger claim's refutation and can be reused in the revised version (the surviving-repertoire role). A revision cycle that simply deletes the original claim and replaces it with the corrected one has completed Stages 1 through 3 but skipped the archival half of Stage 4, discarding exactly the kind of afterlife content this paper's own corpus, cited above, argues is worth designing rather than leaving to accident.

VI. Assessment Criteria

Assessment under this framework requires two changes to a standard rubric: a new graded category, and a re-weighting of an existing one.

Refutation Quality — A New Graded Category

Scored independently of whether the student's own original claim was correct. The three criteria below are adapted, for classroom use, from the three standards Huynh's *Popperian-Holmesian Playbook* specifies for a genuine falsification condition in adult argumentative practice; the adaptation preserves the substance of each standard while restating it in terms a teacher can apply to student work directly.

- **Validity:** does the counterexample or objection actually defeat the claim as stated, or does it attack a different, unstated version of it — a straw-man refutation, scored as a partial failure of the same kind as an invalid proof?
- **Specificity:** does the falsification condition name a checkable observation or a derivable consequence, at a stated threshold distinguishing “the condition is met” from “it is not,” or does it remain at the level of “more evidence would help” — unspecific, and on A1, close to useless? A condition naming the type of evidence required without a threshold for what counts as meeting it fails this criterion in the same way a diagnostic test with no cutoff value fails to diagnose anything.
- **Independence:** can the refutation be evaluated by a classmate who does not accept the claim's own framing, using an ordinary, shared standard of evidence, or does assessing it require first accepting definitions or assumptions the claim itself supplied? A refutation that can only be judged valid or invalid from inside the claim's own vocabulary has not met an independent

standard, and neither, symmetrically, has a defense of the claim that can only be judged sound from inside that same vocabulary.

- Severity: following Popper's own criterion for what makes a test valuable — did the refutation attempt target the claim's least defensible point, or its most defensible one? A severe, unsuccessful refutation attempt is scored higher than a mild, successful one, since a mild refutation may only show the claim was stated carelessly rather than that it was substantively wrong.

Recognizing Ungraded Refutations: Four Common Errors

Not every student-generated objection that has the surface form of a refutation earns credit under the criteria above. The Popperian-Holmesian Playbook catalogs four ways an adult argument's stated exit conditions can fail while appearing to satisfy Popper's criterion; each has a direct classroom analogue a teacher should watch for when applying the Validity and Specificity criteria above.

- The fake exit: a stated falsification condition that no available classroom method could actually check (“I’d be wrong if human nature turned out to be different”). It has the grammar of a falsification condition without the content of one.
- The circular exit: a condition evaluable only by accepting the claim's own definitions, failing Independence above by construction.
- The too-easy exit: a condition so extreme or unlikely that stating it produces the appearance of falsifiability while functionally protecting the claim from any realistic test.
- The self-protective exit: a condition that, if met, would not actually require the student to revise the original claim, because the claim was stated loosely enough to survive its own stated refutation. This is the classroom instance of what Principle 5 is built to close off: an exit clause is not a real exit clause unless meeting it obligates revision.

Re-Weighted Category — Original Claim Accuracy

No longer scored as the primary or sole determinant of the grade, but as one input to the overall cycle score described in Principle 5, weighted comparably to, not subordinate to, Refutation Quality. A rubric that weights Original Claim Accuracy at eighty percent or more of the total grade, as most conventional rubrics implicitly do, is, on this framework, mis-specified regardless of subject matter, since it rewards confirmation over the falsification work A1 identifies as epistemically load-bearing.

A worked example: a tenth-grade physics unit on Newton's first law might grade a lab report not primarily on whether the student's predicted outcome matched the observed one, but on whether the pre-registered falsification condition was specific and checkable; whether the student's own attempt to defeat their prediction, before running the trial, identified a genuine boundary case — friction, air resistance, measurement precision — rather than a vague hedge; and whether the final report correctly characterizes the outcome as corroboration or refutation rather than as simple success or failure.

VII. Comparison With Established Pedagogical Frameworks

Framework	Central Mechanism	What Is Graded	Relation to Falsification-Based Pedagogy
Inquiry-Based Learning	Student-driven questioning and investigation, structured around a driving question	Typically the quality and completeness of the investigation and final explanation	Compatible, not competing. IBL specifies who poses the question and drives the investigation; FBP specifies what the investigation is for (locating a refutation) and how the result is scored. An IBL unit can incorporate Stages 2–3 without altering its own structure.
Problem-Based Learning	Students work in groups on an authentic, ill-structured problem, teacher as facilitator	Typically group process, solution quality, and collaborative skill	Overlaps most with Productive Failure (§II.3): the object is a workable solution to a problem, not falsification of a propositional claim. PBL can incorporate Stages 2–3 for any claim generated mid-process, but does not by default reward refutation of the group's own working solution as a distinct, positively scored act.
Socratic Method	Teacher-led questioning designed to expose contradictions in a student's stated position	Typically not separately graded; a discussion technique rather than an assessment instrument	Closest in spirit — the elenchus is, structurally, a refutation-hunting dialogue. But the questioning is teacher-initiated and teacher-controlled; FBP distributes the Socratic function across the classroom and makes it a graded, reciprocal competency.
Toulmin Argumentation Pedagogy	Students construct arguments with explicit claim–data–warrant–backing–qualifier–rebuttal structure	Typically the completeness and coherence of the constructed argument, rebuttal included	Clearest structural overlap, since rebuttal is an explicit TAP component. Differs in that TAP's rebuttal is typically self-constructed to pre-empt objections and folded into overall argument quality; FBP's Stage 3 is typically adversarial and Refutation Quality is scored as an independent category.

Falsification-Based Pedagogy is not a rival to any of the four traditions above. It is closer to a set of design constraints — the five axioms and five principles — that can be layered onto any of them,

and the paper's contribution is specifying those constraints precisely enough that layering them on is an operational instruction rather than a philosophical gesture. A teacher already running an inquiry-based or problem-based classroom does not need to abandon that structure to adopt this framework; they need to add Stage 2's falsification condition, Stage 3's adversarial search, and §VI's rebalanced rubric to a structure that, in most cases, already exists.

VIII. Objections and Limitations

Objection 1 — Domain Limits. Not every claim worth teaching admits of a clean falsification condition in the Popperian sense. Historical interpretive claims, literary readings, and much of mathematics, whose claims are typically necessary rather than empirically falsifiable, resist Stage 2 in its strict empirical form. Response: the framework does not require empirical falsifiability in Popper's strict demarcation-criterion sense; Stage 2 can be satisfied by a stated logical, textual, or evidentiary condition appropriate to the domain — a counter-reading supported by textual evidence, a proof of inconsistency in mathematics, a primary source that complicates a historical claim — at the cost of some loss of rigor relative to the natural-science case this paper's axioms were drawn from. This is a genuine limitation, not fully resolved here, and is flagged as such.

Objection 2 — Incentive Gaming. Rewarding refutation risks producing students who perform contrarianism, generating superficially plausible but low-quality refutations to collect credit, or who avoid making any claim specific enough to be refuted. Response: this is precisely why Refutation Quality (§VI) is scored on Validity, Specificity, Independence, and Severity rather than merely on the existence of a stated objection, and why the four common errors are named explicitly as ungraded. A low-validity or low-severity refutation should score below a high-quality confirmed claim, not automatically above it. The framework requires, but does not by itself guarantee, careful rubric application and teacher judgment to avoid this failure mode, and this paper's own axioms provide no algorithm for adjudicating every borderline validity call.

Objection 3 — Foundational-Skill Domains. For content where the goal is fluency or automaticity rather than propositional understanding — multiplication tables, vocabulary acquisition, basic decoding in early literacy — a falsification-centered structure may be actively counterproductive, introducing unnecessary friction into skills better acquired through practice and confirmation. Response: the paper does not claim universal applicability across all instructional goals; A4's rejection of the bucket theory is a claim about the growth of conceptual knowledge specifically, and this paper's scope should be read as bounded to claims and concepts, not to procedural or rote-fluency content, where a different and not incompatible instructional logic likely applies.

Objection 4 — Institutional Feasibility. Standardized assessment regimes external to any single classroom grade confirmation-style correct-answer production almost exclusively, creating a structural disincentive for any teacher to adopt a framework whose internal grading logic is not what the external assessment measures. Response: this is a genuine institutional constraint this paper cannot resolve by argument; it is noted as the primary practical barrier to adoption, distinct from any philosophical objection to the framework's internal coherence.

Objection 5 — The Reflexive Turn: Does Training Refutation-Skill Create an Unfalsifiable Self-Belief? A student who becomes skilled at generating counterexamples, and is repeatedly

rewarded for it under Principle 3, can acquire a belief that this paper's own apparatus is built to flag: I am a strong critical thinker, therefore my conclusions are reliable. This belief is a claim like any other, and if the student treats every instance of being persuaded to revise as confirmation of their own skill and every instance of resisting revision as further confirmation of the same skill, the belief has acquired exactly the self-sealing structure Principle 1 exists to catch — immunized not by the student's reasoning about the world, but by the student's reasoning about their own reasoning. Huynh's *The Personal System Reminder* names the general form of this problem for adult epistemic practice as the epistemic immunity claim: a claim about one's own protected status that is, structurally, no more exempt from falsification than any claim about the external world, and that is specifically likely to escape scrutiny because the tools that would catch it are, by construction, pointed outward. Response: this paper's framework closes the gap the same way that source does — not by adding a sixth principle, but by requiring that Principle 1 apply reflexively. A classroom implementing this framework should include, as a standing entry revisited at each reporting period rather than a claim assessed once, the question “what would show that my skill at finding counterexamples has led me to a wrong conclusion?” asked of the student by the student, with a teacher or peer serving the externally-triggered review role Huynh's paper argues a self-administered check cannot reliably perform alone. This is a genuine, non-trivial addition to classroom design, not a rhetorical closing move, and a teacher adopting this framework without it has left exactly the gap this objection describes open.

IX. Conclusion

Popper's asymmetry between confirmation and refutation has been imported into education as a civic philosophy and as a narrow scientific-method teaching technique, and it stands adjacent to at least four further programs — non-coercive Popperian schooling, productive failure, cognitive-conflict conceptual change, and inoculation theory — each of which shares part of its logic without formalizing it as the organizing principle of assessment itself. The five axioms and five principles developed here do not require new philosophical premises Popper did not already supply. What they require, and what this paper has tried to supply, is the operational translation — the axioms into procedures, the procedures into a rubric, the rubric into a comparison against frameworks already in use — that turns “Popper is relevant to education” into a testable, adoptable classroom design.

The reflexive objection in §VIII is the paper's own final application of its own method: a framework that trains students to distrust unfalsifiable confidence should not itself claim unfalsifiable confidence in its own design. Whether the specific weightings and procedures proposed here survive their own Stage 2 and Stage 3 — whether they, too, meet a refutation attempt they cannot survive — is a question this paper leaves, appropriately, open.

References

- Agassi, J. (2014). *The Hazard Called Education: Essays, Reviews, and Dialogues on Education from Forty-Five Years*. Leiden: Brill.
- Chitpin, S. (2016). *Popper's Approach to Education: A Cornerstone of Teaching and Learning*. New York: Routledge.

- Chitpin, S. (2016). Leading School Improvement: Using Popper's Theory of Learning. *Open Review of Educational Research*, 3(1), 190–203.
- Driver, R., Newton, P., & Osborne, J. (2000). Establishing the Norms of Scientific Argumentation in Classrooms. *Science Education*, 84(3), 287–312.
- Dweck, C. S. (2006). *Mindset: The New Psychology of Success*. New York: Random House.
- Erduran, S. (2018). Toulmin's Argument Pattern as a 'Horizon of Possibilities' in the Study of Argumentation in Science Education. *Cultural Studies of Science Education*, 13, 1091–1099.
- Huynh, G. B. (2026). Falsifiability as Cognitive Self-Vaccination: Popper's Criterion, McGuire's Inoculation, and the Asymmetric Deployment of Epistemic Standards.
- Huynh, G. B. (2026). The Personal System Reminder: A Citizen-Built Architecture for Detecting Propaganda, Verifying Genuine Communicators, and Catching One's Own Self-Sealing Drift, Modeled Directly on Anthropic's System Reminder.
- Huynh, G. B. (2026). The Popperian-Holmesian Playbook: How to Construct Genuine Falsification Conditions and Execute Systematic Exit Elimination.
- Huynh, G. B. (2026). The Pre-Registered Afterlife: A Metamethodology of Functional-Transition Design in Falsifiable Frameworks, With Four Historical Cases.
- Kang, S., Scharmann, L. C., & Noh, T. (2004). Reexamining the Role of Cognitive Conflict in Science Concept Learning. *Research in Science Education*, 34, 71–96.
- Kapur, M. (2008). Productive Failure. *Cognition and Instruction*, 26(3), 379–424.
- Kapur, M. (2014). Productive Failure in Learning Math. *Cognitive Science*, 38(5), 1008–1022.
- Kapur, M. (2016). Examining Productive Failure, Productive Success, Unproductive Failure, and Unproductive Success in Learning. *Educational Psychologist*, 51(2), 289–299.
- Lam, C.-M. (2007). Is Popper's Falsificationist Heuristic a Helpful Resource for Developing Critical Thinking? *Educational Philosophy and Theory*, 39(4), 432–448.
- Lam, C.-M. (2013). A Popperian Approach to Education for Open Society. *Educational Philosophy and Theory*, 45(8), 845–859.
- Lam, C.-M. (2017). Confucianism and Critical Rationalism: Friends or Foes? *Educational Philosophy and Theory*, 49(12), 1136–1145.
- McGuire, W. J. (1961). The Effectiveness of Supportive and Refutational Defenses in Immunizing and Restoring Beliefs Against Persuasion. *Sociometry*, 24(2), 184–197.
- McGuire, W. J. (1964). Inducing Resistance to Persuasion: Some Contemporary Approaches. In L. Berkowitz (Ed.), *Advances in Experimental Social Psychology* (Vol. 1, pp. 191–229). New York: Academic Press.
- Pacaci, et al. (2024). Effectiveness of Conceptual Change Strategies in Science Education: A Meta-Analysis. *Journal of Research in Science Teaching*.
- Popper, K. R. (1945). *The Open Society and Its Enemies*. London: Routledge.

- Popper, K. R. (1959). *The Logic of Scientific Discovery*. London: Hutchinson. (Original work published 1934)
- Popper, K. R. (1963). *Conjectures and Refutations: The Growth of Scientific Knowledge*. London: Routledge & Kegan Paul.
- Popper, K. R. (1972). *Objective Knowledge: An Evolutionary Approach*. Oxford: Clarendon Press.
- Popper, K. R. (1983). *Realism and the Aim of Science*. Totowa, NJ: Rowman and Littlefield.
- Posner, G. J., Strike, K. A., Hewson, P. W., & Gertzog, W. A. (1982). Accommodation of a Scientific Conception: Toward a Theory of Conceptual Change. *Science Education*, 66(2), 211–227.
- Sinha, T., & Kapur, M. (2021). When Problem Solving Followed by Instruction Works: Evidence for Productive Failure. *Review of Educational Research*, 91(5), 761–798.
- Taking Children Seriously. *Unschooling and Karl Popper*. Accessed July 2026.
- Taking Children Seriously. *Coercive Education*. Accessed July 2026.
- Toulmin, S. (1958). *The Uses of Argument*. Cambridge: Cambridge University Press.
- Zecha, G. (Ed.). (1999). *Critical Rationalism and Educational Discourse*. Amsterdam: Rodopi.